

# GridBalance — Data Export — Gridbalance Artifact Index

Official Research Data Export — Total Records: 40

| artifact                        | path                                                    | type             | purpose                                                    | research_section | status   |
|---------------------------------|---------------------------------------------------------|------------------|------------------------------------------------------------|------------------|----------|
| SGCC Baseline XGBoost Model     | models/sgcc_baselines/XGBoost_Weighted.joblib           | Model Checkpoint | Baseline XGBoost model with class weighting                | Section 7        | Verified |
| SGCC Baseline SMOTE Model       | models/sgcc_baselines/XGBoost_SMOTE.joblib              | Model Checkpoint | Baseline XGBoost model with SMOTE                          | Section 7        | Verified |
| SGCC Tuned Best Model           | models/sgcc_tuned/xgboost_tuned_best.joblib             | Model Checkpoint | Frozen 42-feature tuned XGBoost classifier                 | Section 9 & 11   | Verified |
| SGCC Reduced Best Model         | models/sgcc_tuned/xgboost_reduced_best.joblib           | Model Checkpoint | Frozen 18-feature candidate XGBoost classifier             | Section 14 & 15  | Verified |
| SGCC Stability Model            | models/sgcc_stability/xgboost_stability_selected.joblib | Model Checkpoint | Stability-selected 14-feature XGBoost classifier           | Section 14       | Verified |
| UCI Forecasting Frozen Model    | models/uci_forecasting/xgb_forecasting_frozen.joblib    | Model Checkpoint | Frozen 29-feature XGBoost regressor for load forecasting   | Section 23 & 24  | Verified |
| SGCC Data Validation Report     | reports/REAL_SGCC_DATA_VALIDATION.md                    | Markdown Report  | Raw SGCC dataset audit and quality metrics                 | Section 3 & 4    | Verified |
| SGCC Feature Engineering Report | reports/REAL_SGCC_FEATURE_ENGINEERING_REPORT.md         | Markdown Report  | Chronological sorting and feature extraction documentation | Section 5        | Verified |
| SGCC Baseline Model Report      | reports/REAL_SGCC_BASELINE_MODEL_REPORT.md              | Markdown Report  | Baseline performance across 12 configurations              | Section 7 & 8    | Verified |
| SGCC XGBoost Tuning Report      | reports/SGCC_XGBOOST_TUNING_REPORT.md                   | Markdown Report  | Randomized CV search and threshold optimization            | Section 9 & 10   | Verified |
| SGCC SHAP Analysis Report       | reports/SGCC_SHAP_FEATURE_ANALYSIS.md                   | Markdown Report  | SHAP feature importances and 18-feature selection          | Section 12 & 13  | Verified |
| SGCC Feature Stability Report   | reports/SGCC_FEATURE_SELECTION_STABILITY.md             | Markdown Report  | Cross-validation stability analysis                        | Section 14       | Verified |
| SGCC Robustness Analysis Report | reports/SGCC_ROBUSTNESS_ANALYSIS.md                     | Markdown Report  | Noise, data loss, and evasion simulation results           | Section 15       | Verified |
| UCI Dataset Audit Report        | reports/UCI_DATASET_AUDIT.md                            | Markdown Report  | Raw UCI load dataset structure and aggregation             | Section 17       | Verified |
| UCI Temporal Split Report       | reports/UCI_TEMPORAL_SPLIT.md                           | Markdown Report  | Train/val/test chronological partition documentation       | Section 19       | Verified |
| UCI Error Analysis Report       | reports/UCI_ERROR_ANALYSIS.md                           | Markdown Report  | Residual and error distribution breakdown                  | Section 25       | Verified |
| UCI Feature Importance Report   | reports/UCI_FEATURE_IMPORTANCE.md                       | Markdown Report  | Feature importance breakdown for UCI regressor             | Section 26       | Verified |
| UCI Reproducibility Report      | reports/UCI_REPRODUCIBILITY.md                          | Markdown Report  | Environment and reproducibility specification              | Section 33       | Verified |
| Backend Inference Report        | reports/ML_INFERENCE_BACKEND_REPORT.md                  | Markdown Report  | FastAPI inference backend architecture and tests           | Section 30       | Verified |

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|----------------------------------|-------------------------------------------------|------------------|--------------------------------------------------|------------------|----------|
| Platform Integration Report      | reports/PLATFORM_INTEGRATION_REPORT.md          | Markdown Report  | Integrated web application documentation         | Section 29 & 31  | Verified |
| Real SGCC Data Quality CSV       | reports/REAL_SGCC_DATA_QUALITY.csv              | CSV Data File    | Raw missingness and zero counts for SGCC         | Section 3 & 4    | Verified |
| Real SGCC Baseline Results CSV   | reports/REAL_SGCC_BASELINE_RESULTS.csv          | CSV Data File    | Empirical metrics for 12 baseline configurations | Section 7        | Verified |
| SGCC Separability Audit CSV      | reports/SGCC_Class_Separability_Audit.csv       | CSV Data File    | Standardized mean differences per feature        | Section 4        | Verified |
| SGCC SHAP Feature Importance CSV | reports/SGCC_SHAP_FEATURE_IMPORTANCE.csv        | CSV Data File    | Mean absolute and signed SHAP values             | Section 12       | Verified |
| SGCC Group Ablation CSV          | reports/SGCC_FEATURE_GROUP_ABLATION.csv         | CSV Data File    | Ablation results for 8 feature groups            | Section 13       | Verified |
| SGCC Selection Frequency CSV     | reports/SGCC_FEATURE_SELECTION_FREQUENCY.csv    | CSV Data File    | Fold selection counts across 5-fold CV           | Section 14       | Verified |
| SGCC Feature Set Comparison CSV  | reports/SGCC_FEATURE_SET_COMPARISON.csv         | CSV Data File    | 42 vs 18 vs 14 feature comparison                | Section 14       | Verified |
| SGCC Noise Perturbation CSV      | reports/SGCC_NOISE_PERTURBATION_RESULTS.csv     | CSV Data File    | Additive and multiplicative noise results        | Section 15       | Verified |
| SGCC Missingness Robustness CSV  | reports/SGCC_MISSINGNESS_ROBUSTNESS_RESULTS.csv | CSV Data File    | Telemetry loss simulation results                | Section 15       | Verified |
| SGCC Evasion Simulation CSV      | reports/SGCC_EVASION_SIMULATION_RESULTS.csv     | CSV Data File    | Load suppression tampering simulation results    | Section 15       | Verified |
| UCI Statistics CSV               | reports/UCI_DATASET_STATISTICS.csv              | CSV Data File    | Summary statistics for UCI dataset               | Section 17       | Verified |
| UCI Model Comparison CSV         | reports/UCI_MODEL_COMPARISON.csv                | CSV Data File    | Validation metrics across forecasting models     | Section 21       | Verified |
| UCI Time Series CV CSV           | reports/UCI_TIME_SERIES_CV.csv                  | CSV Data File    | Expanding window temporal CV results             | Section 22       | Verified |
| FastAPI Main Application         | main.py                                         | Python Script    | FastAPI entry point and lifespan manager         | Section 30       | Verified |
| SGCC Predictor Service           | services/sgcc_service.py                        | Python Module    | SGCC inference service and feature extractor     | Section 30       | Verified |
| UCI Forecaster Service           | services/forecasting_service.py                 | Python Module    | UCI forecasting inference service                | Section 30       | Verified |
| Detection API Router             | api/detection.py                                | Python Module    | End points /api/v1/detection/analyze & batch     | Section 30       | Verified |
| Forecasting API Router           | api/forecasting.py                              | Python Module    | Endpoint /api/v1/forecast/next-hour              | Section 30       | Verified |
| React App Entry Point            | frontend/src/App.tsx                            | TypeScript React | Frontend router and root layout component        | Section 31       | Verified |
| Automated Test Suite             | tests/test_api_endpoints.py                     | Python Pytest    | Backend API endpoint tests                       | Section 32       | Verified |